

Filters and Plots

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Background

In this notebook, we'll review a scenario, and create a data visualization with `ggplot2`. We will make use of the filters and facets features of `ggplot2` to create custom visualizations based on different criteria.

The Scenario

As junior data analysts for a hotel booking company, we have been asked to clean hotel booking data, create visualizations with `ggplot2` to gain insight into the data, and present different facets of the data through visualization. Now, we are going to build on the work we performed previously to apply filters to our data visualizations in `ggplot2`.

Step 1: Import our data

The data in this example is originally from the article Hotel Booking Demand Datasets

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

(<https://www.sciencedirect.com/science/article/pii/S2352340918315191>)), written by Nuno Antonio, Ana Almeida, and Luis Nunes for Data in Brief, Volume 22, February 2019.

The data was downloaded and cleaned by Thomas Mock and Antoine Bichat for #TidyTuesday during the week of February 11th, 2020 (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md> (<https://github.com/rfordatascience/tidytuesday/blob/master/data/2020/2020-02-11/readme.md>)).

We can learn more about the dataset here: <https://www.kaggle.com/jessemostipak/hotel-booking-demand> (<https://www.kaggle.com/jessemostipak/hotel-booking-demand>)

Read in the file 'hotel_bookings.csv' into a data frame:

```
hotel_bookings <- read.csv("hotel_bookings.csv")
```

Step 2: Refresh Our Memory

By now, we are pretty familiar with this data set in our previous notebooks. But we can refresh our memory with the `head()` and `colnames()` functions.

```
head(hotel_bookings)
```

```

##      hotel is_canceled lead_time arrival_date_year arrival_date_month
## 1 Resort Hotel          0       342          2015          July
## 2 Resort Hotel          0       737          2015          July
## 3 Resort Hotel          0         7          2015          July
## 4 Resort Hotel          0        13          2015          July
## 5 Resort Hotel          0        14          2015          July
## 6 Resort Hotel          0        14          2015          July
##  arrival_date_week_number arrival_date_day_of_month stays_in_weekend_nights
## 1              27              1              0
## 2              27              1              0
## 3              27              1              0
## 4              27              1              0
## 5              27              1              0
## 6              27              1              0
##  stays_in_week_nights adults children babies meal country market_segment
## 1              0        2         0      0  BB    PRT    Direct
## 2              0        2         0      0  BB    PRT    Direct
## 3              1        1         0      0  BB    GBR    Direct
## 4              1        1         0      0  BB    GBR    Corporate
## 5              2        2         0      0  BB    GBR    Online TA
## 6              2        2         0      0  BB    GBR    Online TA
##  distribution_channel is_repeated_guest previous_cancellations
## 1          Direct              0              0
## 2          Direct              0              0
## 3          Direct              0              0
## 4      Corporate              0              0
## 5          TA/T0              0              0
## 6          TA/T0              0              0
##  previous_bookings_not_canceled reserved_room_type assigned_room_type
## 1              0              C              C
## 2              0              C              C
## 3              0              A              C
## 4              0              A              A
## 5              0              A              A
## 6              0              A              A
##  booking_changes deposit_type agent company days_in_waiting_list customer_type
## 1              3  No Deposit  NULL  NULL              0  Transient
## 2              4  No Deposit  NULL  NULL              0  Transient
## 3              0  No Deposit  NULL  NULL              0  Transient
## 4              0  No Deposit  304  NULL              0  Transient
## 5              0  No Deposit  240  NULL              0  Transient
## 6              0  No Deposit  240  NULL              0  Transient
##  adr required_car_parking_spaces total_of_special_requests reservation_status
## 1  0              0              0  Check-Out
## 2  0              0              0  Check-Out
## 3  75              0              0  Check-Out
## 4  75              0              0  Check-Out
## 5  98              0              1  Check-Out
## 6  98              0              1  Check-Out
##  reservation_status_date
## 1      2015-07-01
## 2      2015-07-01

```

```
## 3      2015-07-02
## 4      2015-07-02
## 5      2015-07-03
## 6      2015-07-03
```

```
colnames(hotel_bookings)
```

```
## [1] "hotel" "is_canceled"
## [3] "lead_time" "arrival_date_year"
## [5] "arrival_date_month" "arrival_date_week_number"
## [7] "arrival_date_day_of_month" "stays_in_weekend_nights"
## [9] "stays_in_week_nights" "adults"
## [11] "children" "babies"
## [13] "meal" "country"
## [15] "market_segment" "distribution_channel"
## [17] "is_repeated_guest" "previous_cancellations"
## [19] "previous_bookings_not_canceled" "reserved_room_type"
## [21] "assigned_room_type" "booking_changes"
## [23] "deposit_type" "agent"
## [25] "company" "days_in_waiting_list"
## [27] "customer_type" "adr"
## [29] "required_car_parking_spaces" "total_of_special_requests"
## [31] "reservation_status" "reservation_status_date"
```

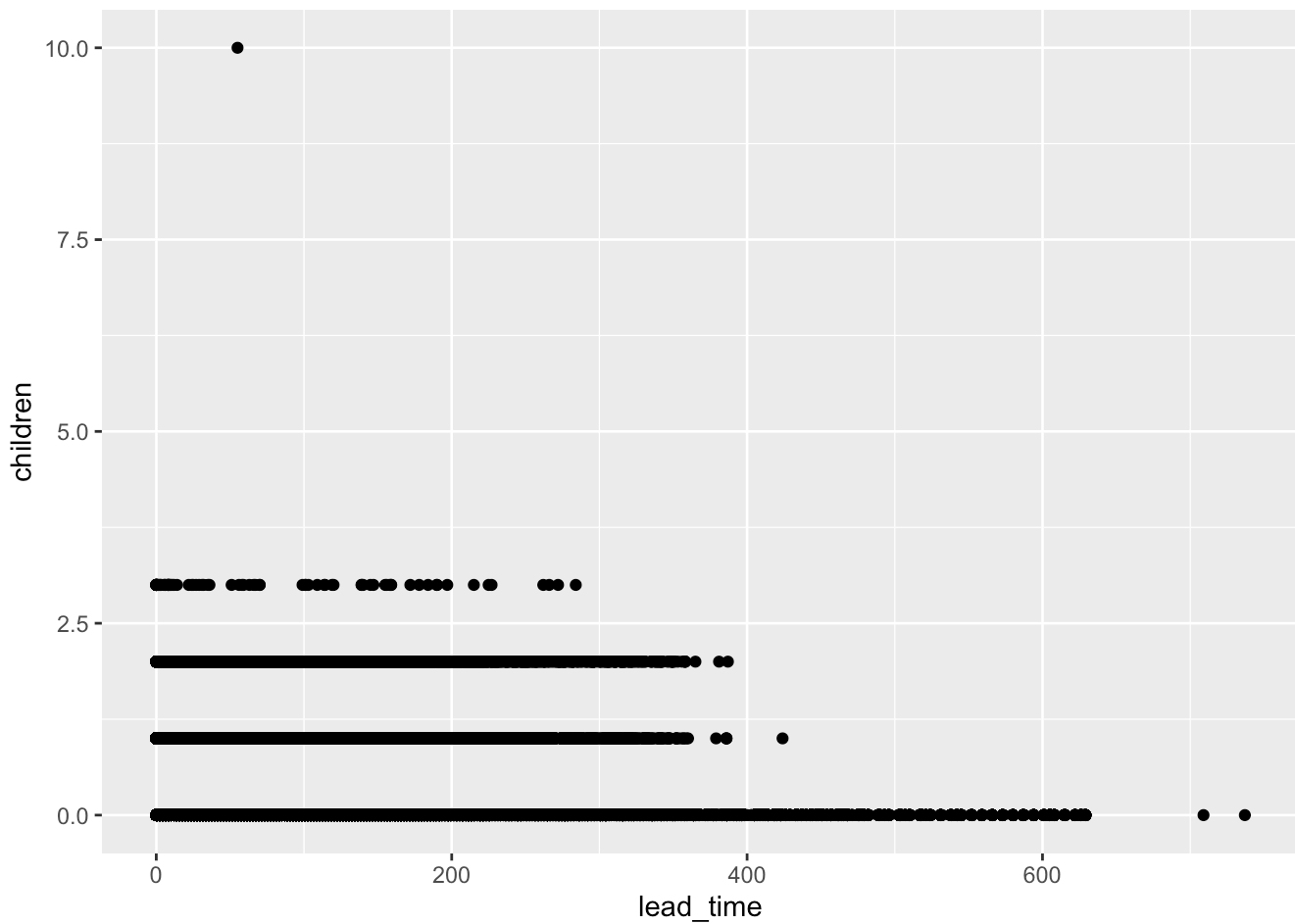
Step 3: Install and load the ‘ggplot2’ package

Step 4: Making many different charts

As a refresher, here’s the scatter plot we created earlier.

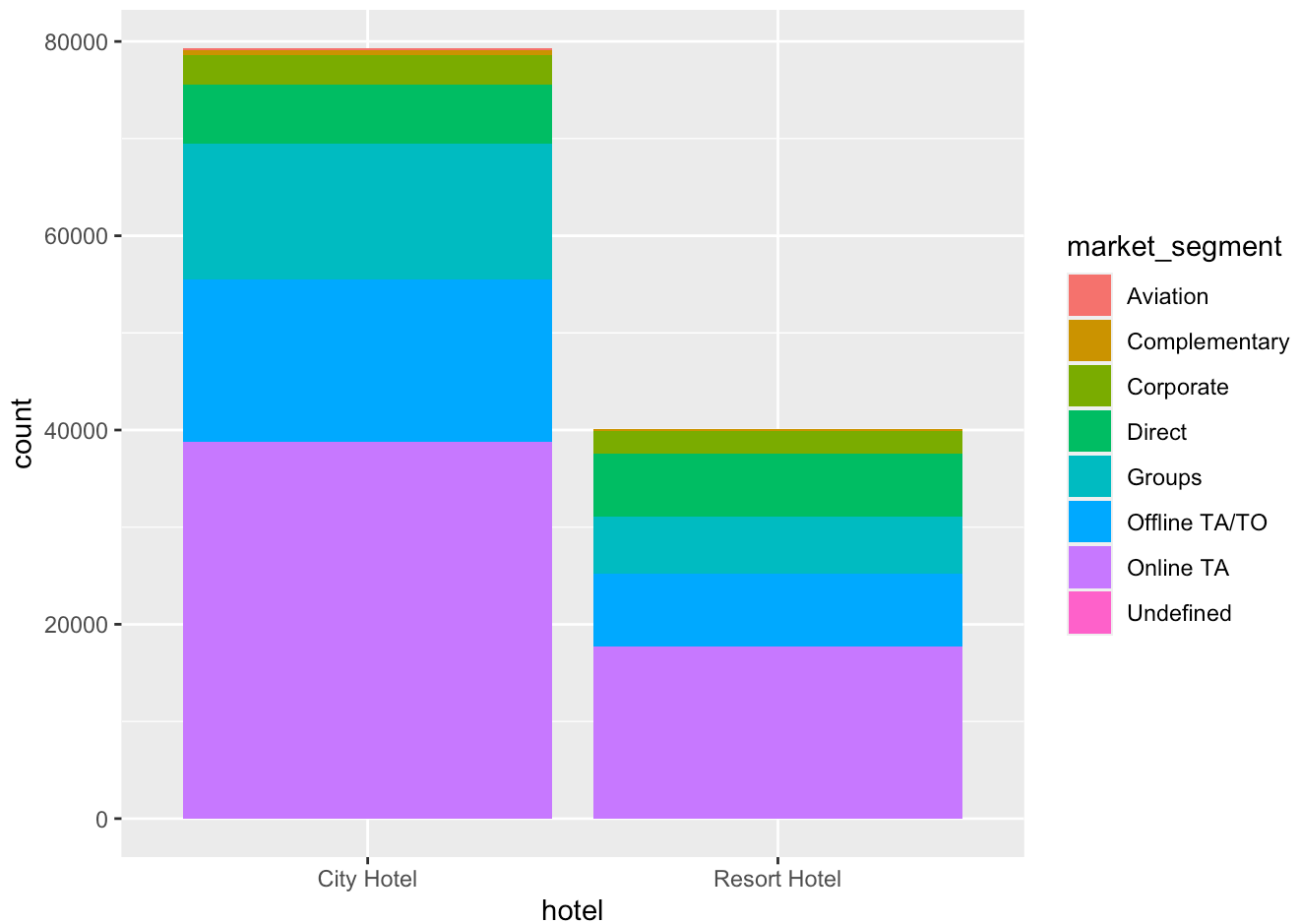
```
ggplot(data = hotel_bookings) +
  geom_point(mapping = aes(x = lead_time, y = children))
```

```
## Warning: Removed 4 rows containing missing values (`geom_point()`).
```



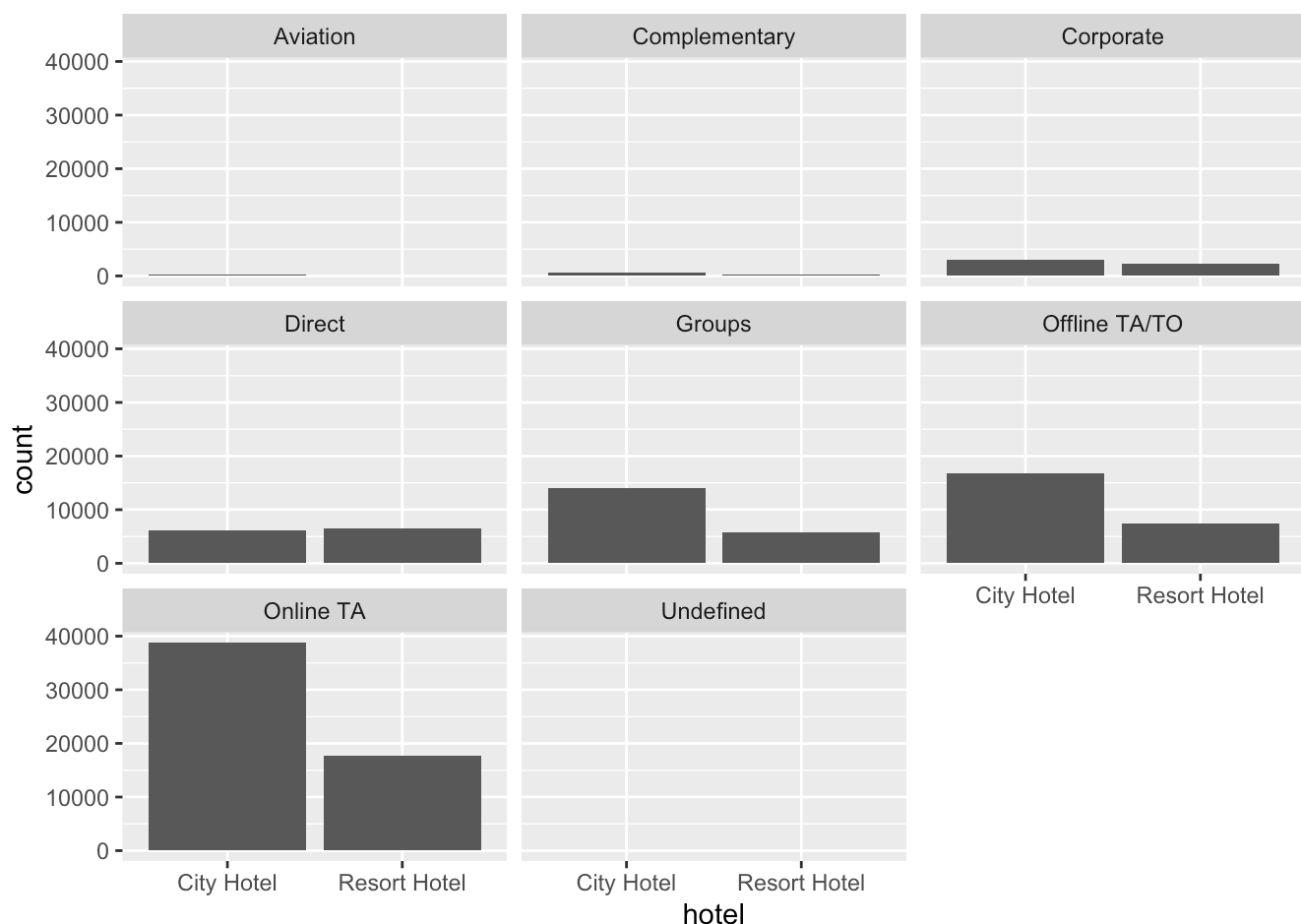
We decide to create a bar chart showing each hotel type and market segment. We use different colors to represent each market segment:

```
ggplot(data = hotel_bookings) +  
  geom_bar(mapping = aes(x = hotel, fill = market_segment))
```



We decide to use the `facet_wrap()` function to create a separate plot for each market segment:

```
ggplot(data = hotel_bookings) +  
  geom_bar(mapping = aes(x = hotel)) +  
  facet_wrap(~market_segment)
```



Step 5: Filtering

For the next step, we will need to have the `tidyverse` package installed and loaded.

```
#install.packages('tidyverse')
library(tidyverse)
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.4
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ lubridate  1.9.3      ✓ tibble     3.2.1
## ✓ purrr      1.0.2      ✓ tidyr      1.3.0
## — Conflicts ————— tidyverse_conflicts() —
## * dplyr::filter() masks stats::filter()
## * dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts
to become errors
```

Use the `filter()` function to create a data set that only includes the data we want:

```
onlineta_city_hotels <- filter(hotel_bookings,
                              (hotel=="City Hotel" &
                               hotel_bookings$market_segment=="Online TA"))
```

We can use the `View()` function to check out our new data frame:

```
#View(onlineta_city_hotels)
```

We can use the pipe operator (`%>%`) to do this in steps!

We name this data frame `onlineta_city_hotels_v2`:

```
onlineta_city_hotels_v2 <- hotel_bookings %>%  
  filter(hotel=="City Hotel") %>%  
  filter(market_segment=="Online TA")
```

This code chunk generates the same data frame by using the `View()` function:

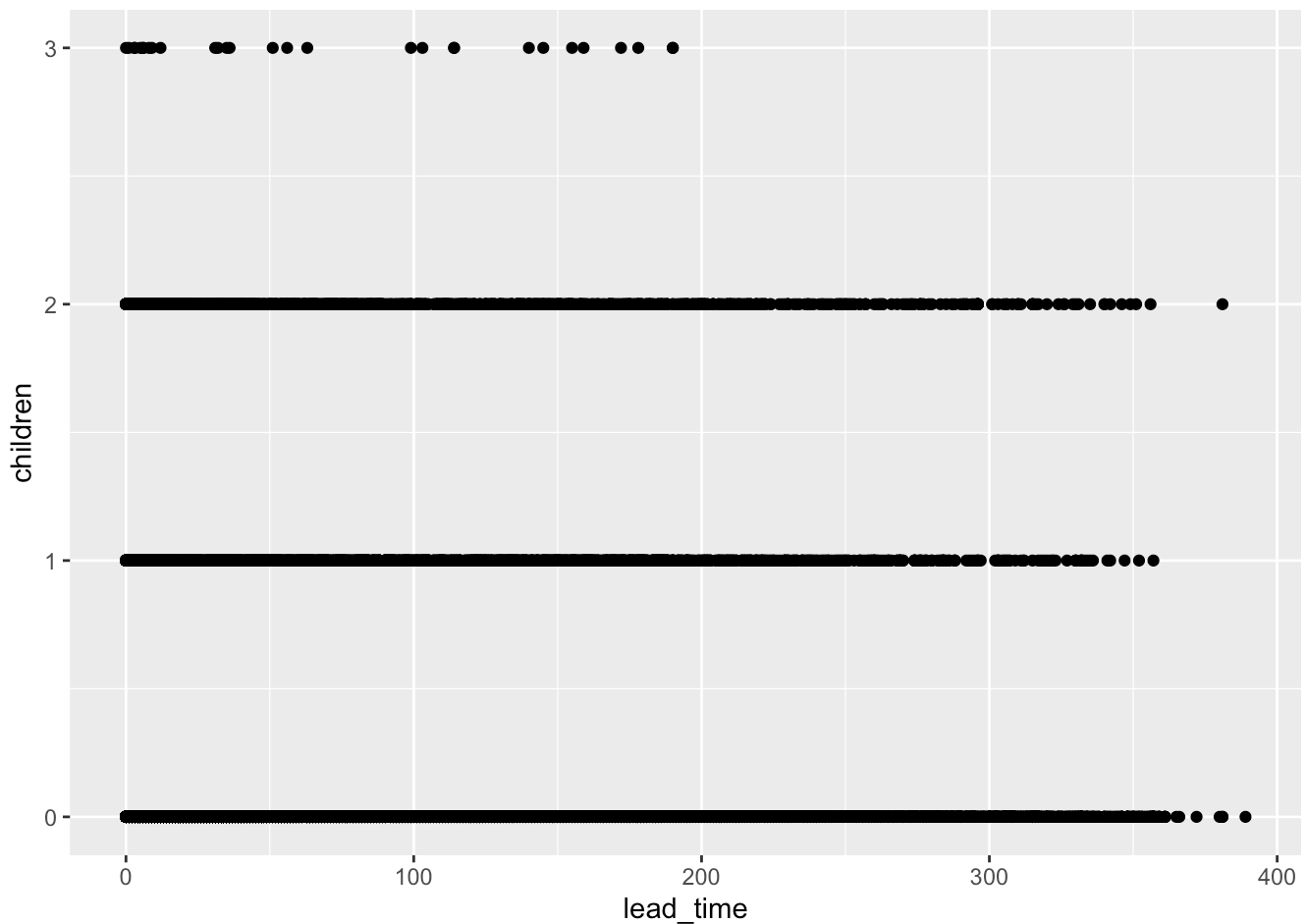
```
#View(onlineta_city_hotels_v2)
```

Step 6: Use our new dataframe

Using the code for our previous scatterplot, replace `data` with either `onlineta_city_hotels` to plot the data our stakeholder requested:

```
ggplot(data = onlineta_city_hotels) +  
  geom_point(mapping = aes(x = lead_time, y = children))
```

```
## Warning: Removed 1 rows containing missing values (`geom_point()`).
```



Based on our previous filter, this scatterplot shows data for online bookings for city hotels. The plot reveals that bookings with children tend to have a shorter lead time, and bookings with 3 children have a significantly shorter lead time (<200 days). So, promotions targeting families can be made closer to the valid booking dates.

Wrap Up

Filters allow us to create different views of our data and allow us to investigate more specific relationships within our data.